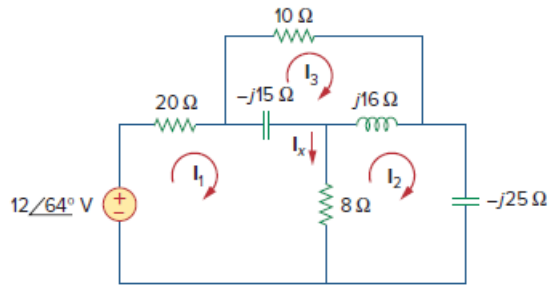


Problem

Find I_1 , I_2 , I_3 , and I_x in the circuit of Fig. 10.84.

Figure 10.84:



Step-by-step solution

Step 1 of 4

For Mesh 1,

$$(28 - j15)I_1 - 8I_2 + j15I_3 = 12\angle 64^\circ \dots\dots\dots (1)$$

For Mesh 2,

$$-8I_1 + (8 - j9)I_2 - j16I_3 = 0 \dots\dots\dots (2)$$

For Mesh 3,

$$j15I_1 - j16I_2 + (10 + j)I_3 = 0 \dots\dots\dots (3)$$

Step 2 of 4

In matrix form (1), (2) & (3) can be cast as or $[A]I = B$

$$\begin{bmatrix} 28 - j15 & -8 & j15 \\ -8 & (8 - j9) & -j16 \\ j15 & -j16 & 10 + j \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 12\angle 64^\circ \\ 0 \\ 0 \end{bmatrix}$$

Step 3 of 4

$$\Delta = \begin{bmatrix} 28-j15 & -8 & j15 \\ -8 & (8-j9) & -j16 \\ j15 & -j16 & 10+j \end{bmatrix}$$

$$\Delta = 5750 - j9560$$

$$\Delta_1 = \begin{bmatrix} 5.26 + j10.78 & -8 & j15 \\ 0 & (8-j9) & -j16 \\ 0 & -j16 & 10+j \end{bmatrix}$$

$$\Delta_1 = 2698.7 + j3287.8$$

$$\Delta_2 = \begin{bmatrix} 28-j15 & 5.26 + j10.78 & j15 \\ -8 & 0 & -j16 \\ j15 & 0 & 10+j \end{bmatrix}$$

$$\Delta_2 = 1597 + j3491.7$$

$$\Delta_3 = \begin{bmatrix} 28-j15 & -8 & 5.26 + j10.78 \\ -8 & (8-j9) & 0 \\ j15 & -j16 & 0 \end{bmatrix}$$

$$\Delta_3 = -796.34 - j1413.2$$

Step 4 of 4

Therefore,

$$I_1 = \frac{\Delta_1}{\Delta}$$

$$I_1 = \frac{2698.7 + j3287.8}{5750 - j9560}$$

$$\therefore I_1 = 0.3814 \angle 109.6^\circ \text{ A}$$

$$I_2 = \frac{\Delta_2}{\Delta}$$

$$I_2 = \frac{1597 + j3491.7}{5750 - j9560}$$

$$\therefore I_2 = 0.3443 \angle 124.4^\circ \text{ A}$$

$$I_3 = \frac{\Delta_3}{\Delta}$$

$$I_3 = \frac{-796.34 - j1413.2}{5750 - j9560}$$

$$\therefore I_3 = 0.145 \angle -60.42^\circ \text{ A} \text{ and}$$

$$I_X = I_1 - I_2$$

$$\therefore I_X = 0.10045 \angle 48.5^\circ \text{ A}$$